

Retail Sponsor: Customer Segmentation & Shelf Space Optimization

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[Names Redacted]



Agenda

-  Why Shelf Space Matters?
-  Business Value & Objectives
-  Data Assets: Structure & Breakdown
-  Analytical Approach
-  Next Steps & Recommendations



Why Shelf Space Strategy Matters?



A single planogram for ~2,000 stores leaves significant revenue on the table

Space Misalignment

- Over/under allocation vs. Actual demand
- Lost revenue at store level

Silo Management

- Categories managed independently
- Miss cross-item (halo/cannibalization) effects

No Scalability

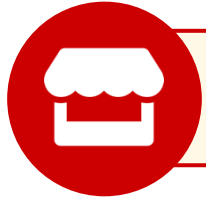
- One-size-fits-all planogram
- Does not adapt across ~2,000 stores



Business Value



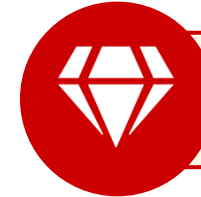
Business Outcomes



Maximize store
efficiencies

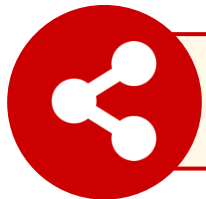


Better customer
satisfaction



Increase Revenue &
Reduce Costs

Our Analytical Approach



Clustering Stores
by Demographics



Isolated Key Attributes
by Stores



Optimizing Shelf Space
by Brand and Items



Objectives



What Drives Demand?

Identify which store attributes influence demand across item groups and categories.



How Do Drivers Interact?

Assess whether demand is driven by single factors or combinations and how this varies across stores.



Where is Space Misaligned?

Pinpoint where current shelf allocation over or under indexes relative to actual sales share by brand and item type.



How do we test and measure impact?

Define pilot plans, success metrics, and guardrails before full rollout.



How do we scale decisions across stores?

Group stores into 6 to 7 segments and assign clear, repeatable space strategies.



Data Assets & Structure



2024 Revenue by Category		
Baby Care	Health & Immune	Meal Essentials
[Redacted]	[Redacted]	[Redacted]



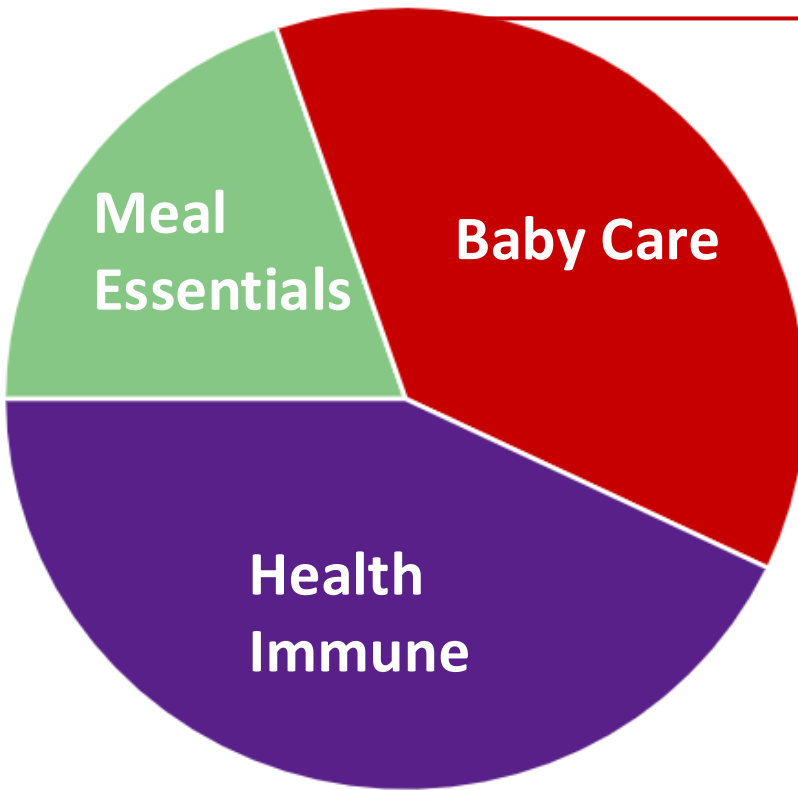
Top 5 Item Types in Baby Care	
Item Type	Sales
Disposable Diapers	[Redacted]
Formula	[Redacted]
Baby Wipes	[Redacted]
Disposable Training Pants	[Redacted]
Baby Snacks	[Redacted]



Top Brands in Item Type Disposable Diapers	
Brands	Sales
[Brand A]	[Redacted]
[Brand B]	[Redacted]
[Brand C]	[Redacted]
[Brand D]	[Redacted]
[Brand E]	[Redacted]



Breakdown/Analytical Framework



- LASSO Regression: ***Feature Selection*** for relevant attributes for Baby Care
- K Means Clustering: ***Customer Segmentation*** dividing into 6 individual clusters by relevant attributes
- Misalignment & Halo/Cannibalization (ElasticNet): ***Optimal Shelf Space Metric*** used per brand x Item Type
- Optimization (Gurobi): *Utilizing metrics to make optimize shelf space*

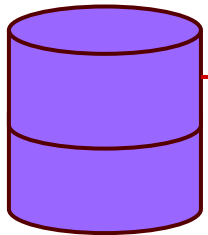


Approach: LASSO Regression



Original Features

70 Census
1 Curbside
1 Facings
1 Format
2 Brand/Item Type
1 Climate
8 Competitor Distance
3 Amex/Snap/Curbside %



*Is demand driven by products
or store attributes?*



Feature Expansion

70 Census



1 Curbside



1 Facings



5 Format



304 Brands/Item Types



1 Climate



8 Competitor Distance



3 Amex/Snap/Curbside %

393 Total Features

LASSO Regression

$$\min \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j * x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

$$\% \text{ of Store Sales} = \beta_0 + \beta_1 \times \text{Feature}_1 + \dots + \beta_n \times \text{Feature}_n$$

$R^2 = 0.861$

Surviving Features

95 Brands
41 Item Types
1 Facings

3 Census

Elastic Net

K-Means
Clustering

Approach: K-Means



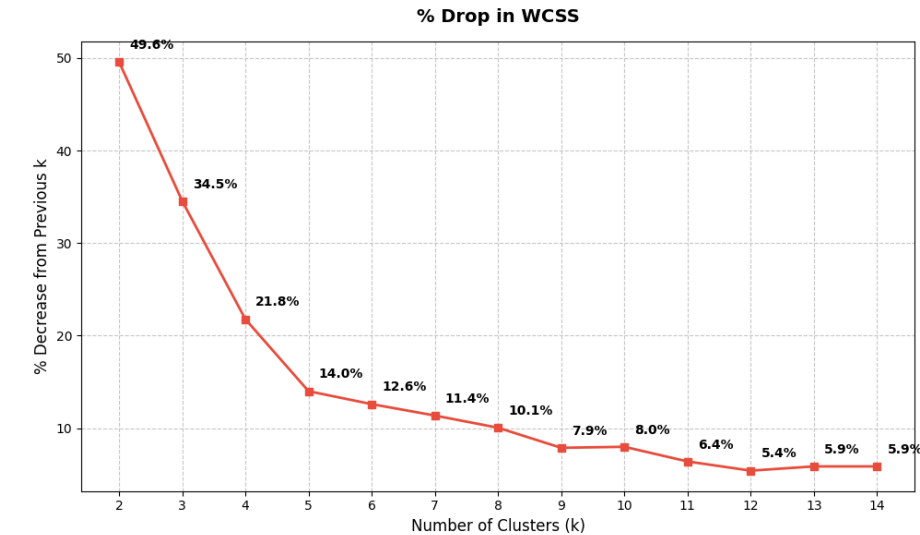
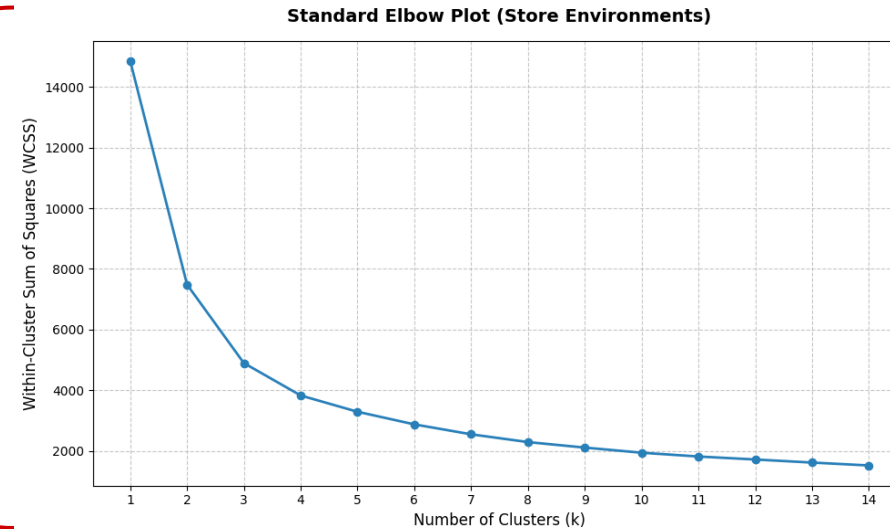
K-Means Clustering

3 Census

Deduplication + Scaling

$$WCSS = \sum_{i=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

Customer
Segmentation



Cluster_ID	1	2	3	4	5	6
Store_Count	710	263	475	47	392	90
news_informed_tv_i	103.31	95	99.98	89.85	97.23	87.67
x_multi_20_plus	5%	20%	9%	76%	7%	36%
somewhat_liberal_index	88.74	132.32	114.74	161.87	99.26	164.68

Cluster 1 Profile

1. Consumes information on average through traditional media
2. Gen X or older Millennial
3. High Homeownership
4. Leans slightly or moderately conservative



Geographic View of Store Cluster

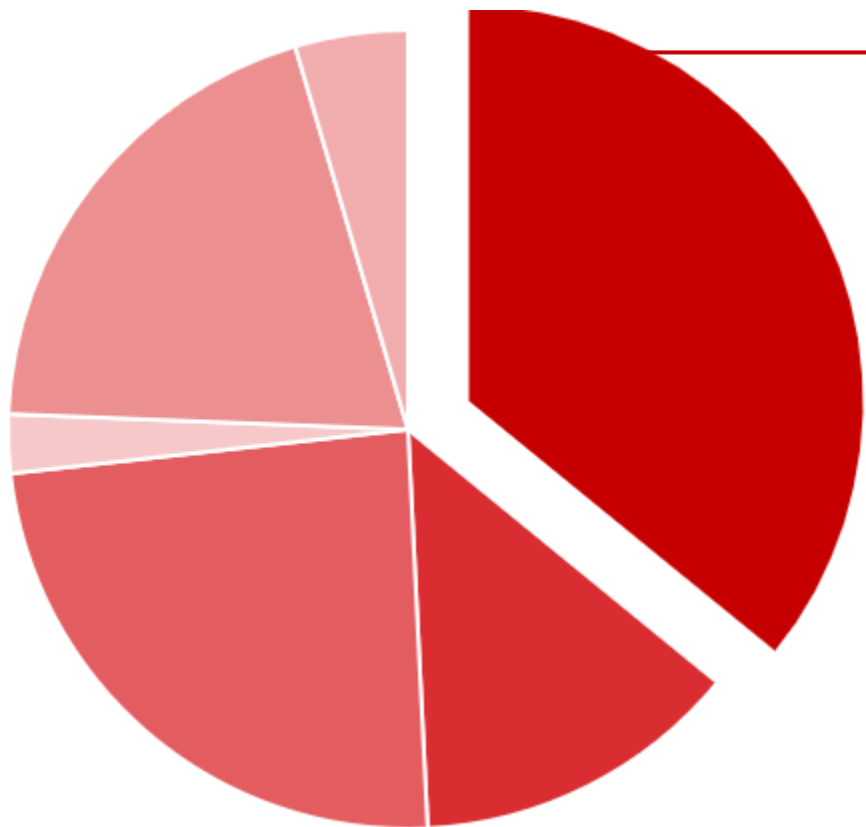


Example: Cluster 1

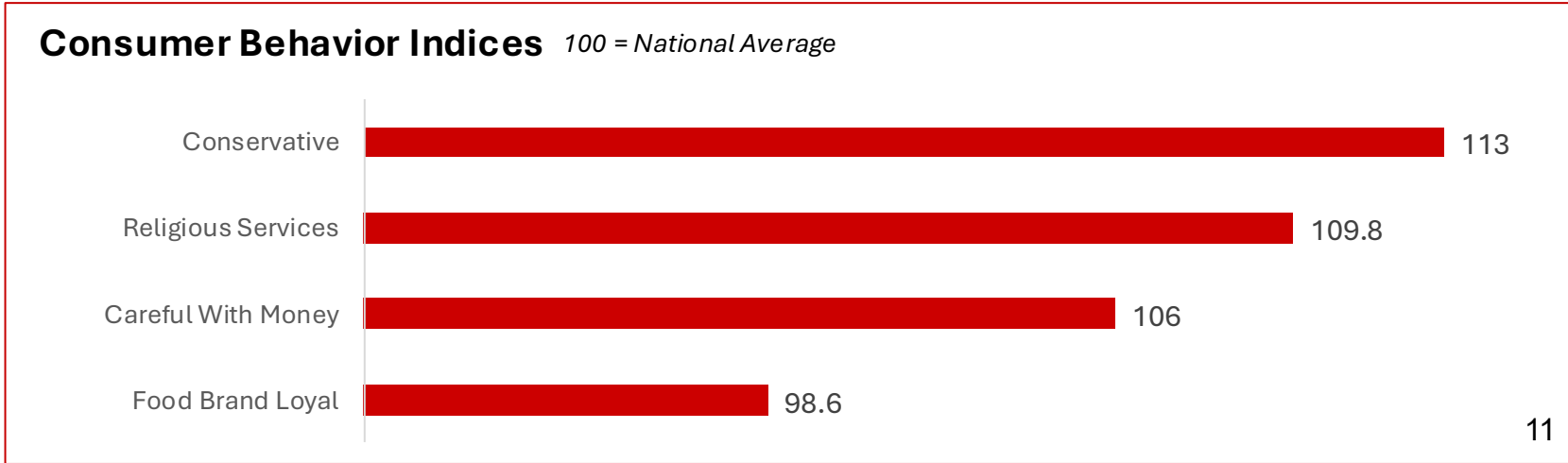
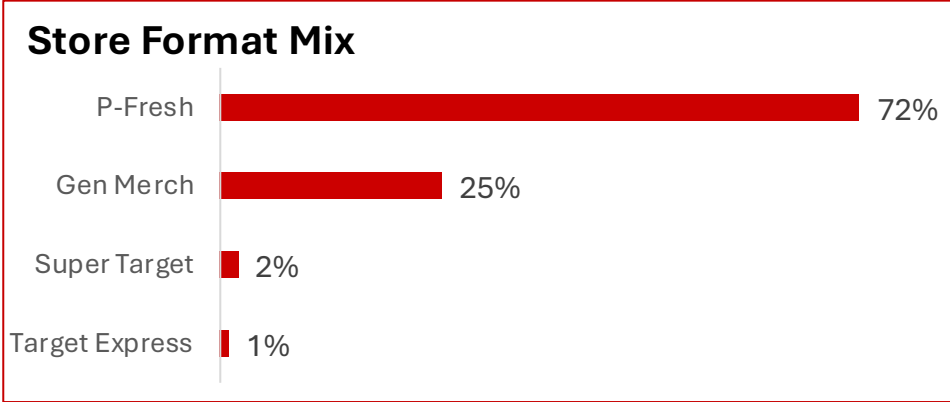
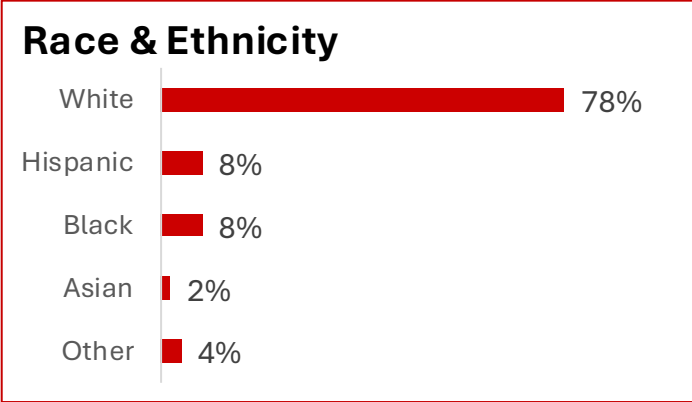


Example

Baby Care - Cluster 1



Median HH Income: [Redacted]	Pop. Density (<i>Sqr. mi</i>): [Redacted]	Share Age (55+): [Redacted]	HH with Kids: [Redacted]
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Approach: Elastic Net



Elastic Net (L1 + L2) Engine

95 Brands + 41 Item Types + Facings

Sparsity Filter (<5% of stores) + Sales Threshold (< 0.1% sales)

$$\min \sum (y - X\beta)^2 + \lambda [\alpha|\beta| + (1 - \alpha)\beta^2]$$

DIRECT EFFECT

Brand's own facing share
change → its own sales

HALO EFFECT

Growing brand A's space
→ lifts brand B's sales

CANNIBALIZATION

Growing brand A's space
→ steals sales from B

Reliability Check ↓

1

Separate EN per item type

2

Coefficient of Variance

3

Beta vs CV

4

VIF Check

5

Trust Score

$$\text{New Sales\%}(\text{Retail Sponsor Brand}) = \text{Current Sales\%}(\text{Retail Sponsor Brand}) + \beta(\text{Lever} \rightarrow \text{Retail Sponsor Brand}) \times \Delta \text{Facing\%}(\text{Lever Brand})$$

CV of Facing % — Within-Category Levers (Higher CV = More Natural Variation = Better Signal)

Lever	Std Dev (facing %)	Mean (facing %)	CV (%)	CV Flag	Item Type
Happy_Hues+Disposable_Training_Pants				HIGH CV — natural variation, model learns	Disposable Training Pants
We_Are_Paradoxx_Mini_Humans+Baby_Lotions				HIGH CV — natural variation, model learns	Baby Lotions
We_Are_Paradoxx_Mini_Humans+Baby_Bath_Wash				HIGH CV — natural variation, model learns	Baby Bath Wash

[Redacted]



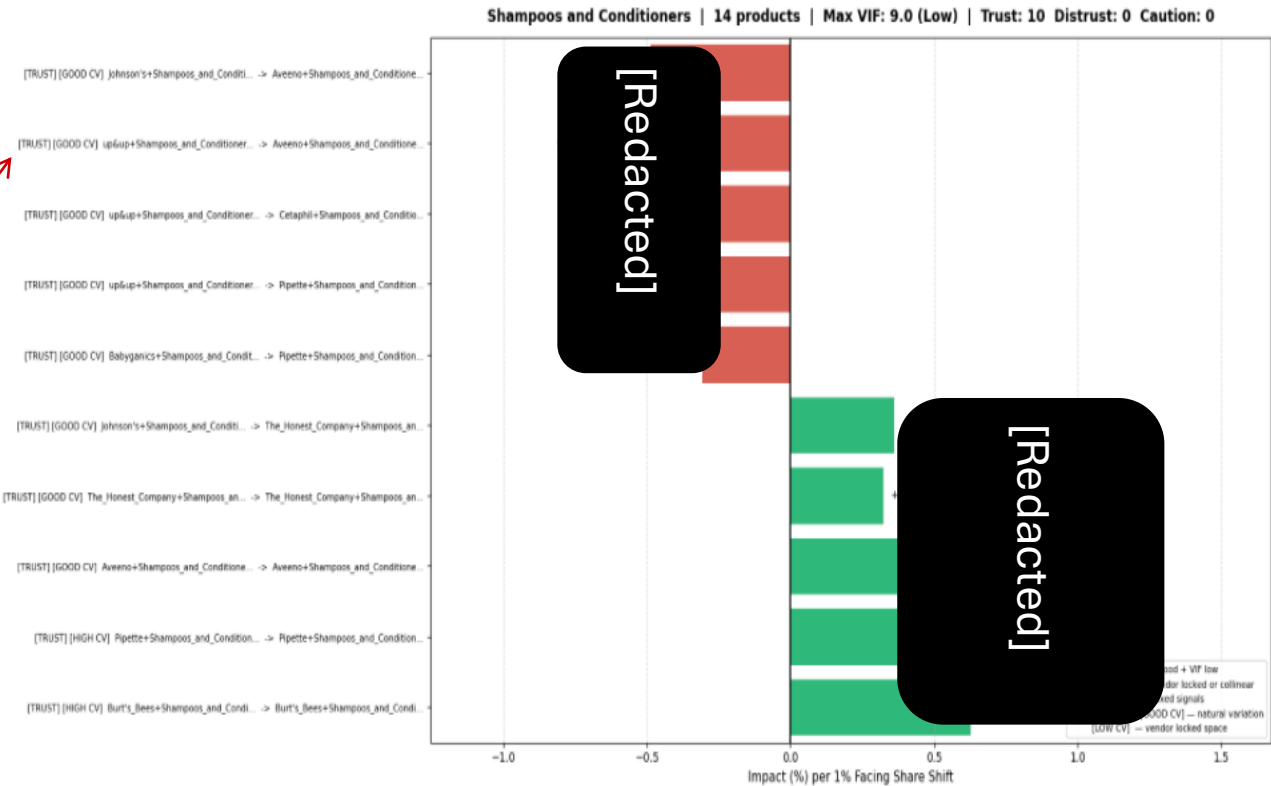
Example: Cannibalization/Halo



Example

- Shampoos and Conditioners:
 - A 1% facing-share increase is associated with about a xx% sales increase
 - Self effect or cross-product effect

Lever	Retail Sponsor	Beta	Interpretation
[Brand C]	[Brand F]	[Redacted]	[Brand C] space steals from [Brand F] (Cannibalization)
[Brand G]	[Brand E]	[Redacted]	[Brand G] space lifts [Brand E] sales (Halo)
[Brand F]	[Brand F]	[Redacted]	Own space → own sales (Direct)



Approach: Shelf Space Misalignment



A facing is one visible unit of a product on the shelf

More facings increase visibility, reduce stockouts, and improve conversion directly impacting sales.

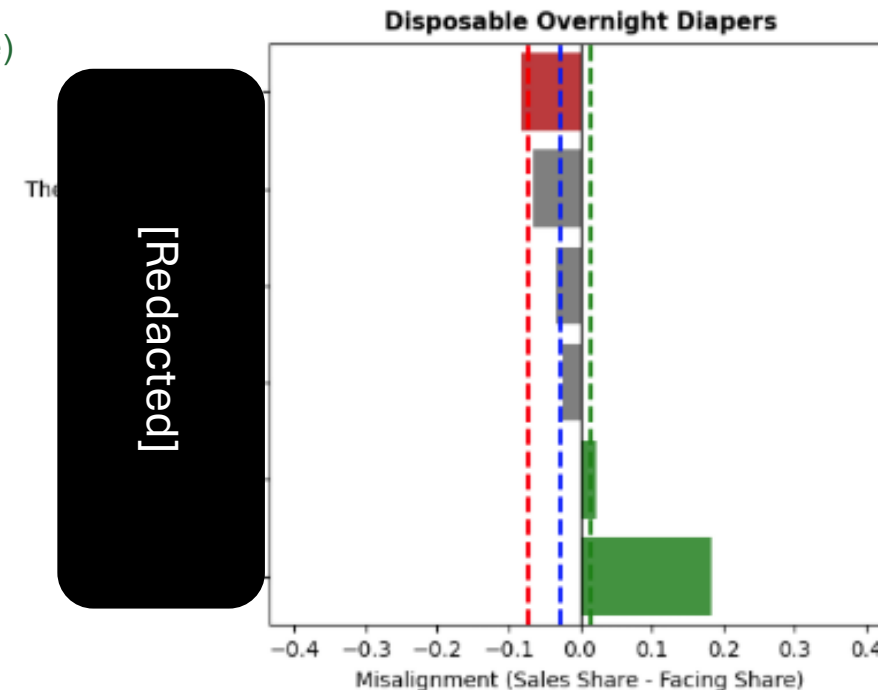
Misalignment = Sales Share – Facing Share

Thresholds = Median $\pm$ (k $\times$ MAD)

MAD (Median Absolute Deviation) is used instead of standard deviation because it is robust to outliers and better reflects typical variation within each category.

Positive = under spaced (needs more space)

Negative = over spaced (too much space)



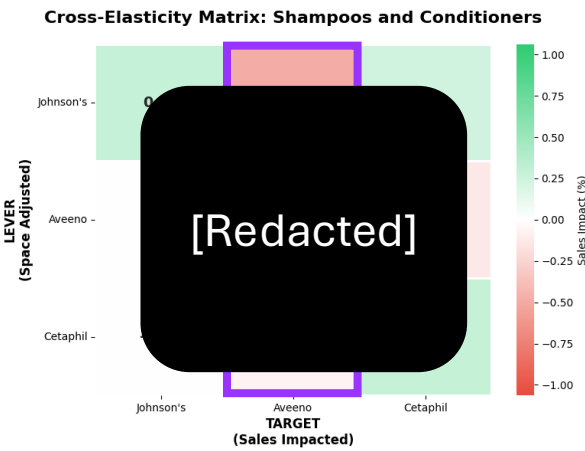
Approach: Gurobi



Elastic Net Betas

Lever	Retail Sponsor	Beta
[Brand F]	[Brand F]	[Redacted]
[Brand G]	[Brand E]	
[Brand C]	[Brand F]	

Convert into cross-elasticity matrix



Gurobi

Objective Function – Maximize Sales

$$\max Z = \sum_t \left[S_t + \sum_d \beta_{d,t} (x_d - F_d) \right]$$

$$\text{New Sales \%} = \text{Current Sales \%} + \beta \times (\text{New Facing\%} - \text{Old Facing\%})$$

Optimized Sales % [Brand F]

$$\begin{aligned} &= \text{Current Sales \% [Brand F]} \\ &- [\text{Redacted}] \times \Delta \text{Cetaphil Facing \%} \\ &+ [\text{Redacted}] \times \Delta [\text{Brand F}] \text{ Facing \%} \\ &- [\text{Redacted}] \times \Delta [\text{Brand G}] \text{ Facing \%} \\ &+ \dots \\ &+ \beta_n \times \Delta \text{Brand}_n \text{ Facing \%} \end{aligned}$$

Constraints

1. Item Type capacity must remain the same
2. No brand can exceed +/- 3% change in capacity
3. Facing % cannot drop below 0%

Optimized Sales %

$$\begin{aligned} &= \text{Optimized Sales \% [Brand F]} \\ &+ \text{Optimized Sales \% [Brand G]} \\ &+ \text{Optimized Sales \% Cetaphil} \\ &+ \dots \\ &+ \text{Optimized Sales \% Brand}_n \end{aligned}$$

Approach: Pearson Correlation



Competitor Distance Data

Apply 8 competitor distance columns to Cleared Item Types only after optimization

Cluster Competition Profile (710 stores, nearest competitor across all 8)

- HIGH (≤3mi) **691 stores (97%)**
- MEDIUM (3–10mi) **16 stores (2%)**
- LOW (>10mi) **3 stores (1%)**

Competitor Proximity — Cluster 1 Stores (shorter distance = more competitive pressure)

Competitor	Mean Dist (mi)	Min Dist (mi)	Max Dist (mi)	Stores < 3mi	Stores < 5mi
F	1.6	0.1	24.9	613	682
C	4.9	0.1	122.4	498	587
E	6.0	0.0	160.9	454	547
H	8.2	0.0	143.3	324	422
A	9.2	0.0	213.8	342	412
G	39.2	0.0	1368.0	26	61
D	227.6	0.1	3624.7	143	156
B	799.9	0.2	3350.5	16	17

Pearson Correlation Test

$$r = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sqrt{[\sum(X_i - \bar{X})^2 \times \sum(Y_i - \bar{Y})^2]}}$$

X = competitor_f distance (miles)
Y = brand_pog_sales \$ per store
N = 710 stores | threshold: $|r| > 0.10$, $p < 0.05$

✓ **$r < -0.10$, $p < 0.05$**

Sales ↑ when rival is CLOSER
→ Brand beats rival head-to-head
→ Retail Sponsor has competitive strength

— **$|r| \leq 0.10$ or $p \geq 0.05$**

No consistent distance pattern
→ Competition not main driver
→ Demographics / price dominates

⚠ **$r > +0.10$, $p < 0.05$**

Sales ↓ when rival is CLOSER
→ Rival takes customers when nearby
→ Cross-shopping risk
→ Caution on shelf expansion



Approach: Pearson Correlation



Expand Pearson correlation analysis across all eight competitors (Brand Sales vs. Distance)

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CATEGORY: DISPOSABLE TRAINING PANTS
Total Cluster Sales: \$226.80M
Signal summary: 15 strong 6 weak 35 neutral

=====

Disposable Training Pants — r value per competitor | Total: \$226.80M | gains closer (strong vs rival) | loses closer (rival steals customers) | no pattern

Brand	Avg Sales % of Cat	vs A	vs B	vs C	vs D	vs E	vs F	vs G	vs H
[Redacted]	0.04%	-0.238	+0.158	-0.057	-0.053	-0.082	-0.036	0.003	-0.150
	0.04%	-0.203	+0.101	-0.028	-0.050	-0.053	-0.037	0.004	-0.136
	0.03%	-0.214	+0.182	-0.054	0.031	-0.040	0.008	-0.023	-0.114
	0.03%	-0.249	-0.011	-0.139	-0.037	-0.137	-0.018	-0.052	-0.211
	0.01%	-0.092	0.002	-0.058	+0.187	-0.120	-0.118	0.036	-0.160
	0.00%	-0.143	0.071	-0.056	+0.266	-0.042	-0.079	+0.180	-0.146
	0.00%	-0.191	0.029	-0.099	-0.052	-0.138	-0.009	0.043	0.010

----- Optimization Action — Final Recommendation Per Brand -----

GROW — Full Confidence

Gurobi shift + signal both agree

- › Gurobi: increase space
- › $r < -0.10 \rightarrow$ beats rival nearby
- › $VIF \leq 10 + CV \geq 10\% \rightarrow$ beta trusted
- › $\rightarrow$ [Brand A], [Brand I] vs rival A

GROW — Rival Risk

Gurobi shift + but signal disagrees

- › Gurobi: increase space
- › $r > +0.10 \rightarrow$ rival steals customers nearby
- › $\rightarrow$ Do NOT execute without review

GROW — Standard

Gurobi shift + signal (neutral)

- › Gurobi: increase space
- › No significant competitive signal
- › Apply shift — monitor proximity
- › $\rightarrow$ [Brand G], [Brand C]

REDUCE — Rival Risk

Gurobi $\downarrow$ + rival compounds loss

- › Gurobi reduces space
- › $r > +0.10 \rightarrow$ rival also hurts brand
- › $\rightarrow$ Double pressure — cut space first
- › $\rightarrow$ Flag for store-level review

REDUCE — Over-Spaced

Gurobi $\downarrow$ but brand still strong

- › Gurobi reduces space
- › Brand still beats rival nearby
- › $\rightarrow$ [Brand H], [Brand A]

REDUCE — Standard

Gurobi $\downarrow$ + no signal

- › Gurobi reduces space
- › No competitive factor
- › $\rightarrow$ [Brand L], [Brand E]



Approach: Pearson Correlation



Example

Pearson r : brand sales vs. distance from each competitor · 710 stores · $|r| > 0.10$, $p < 0.05$ | Retail Sponsor stronger when rival closer | Retail Sponsor weaker when rival closer | no pattern

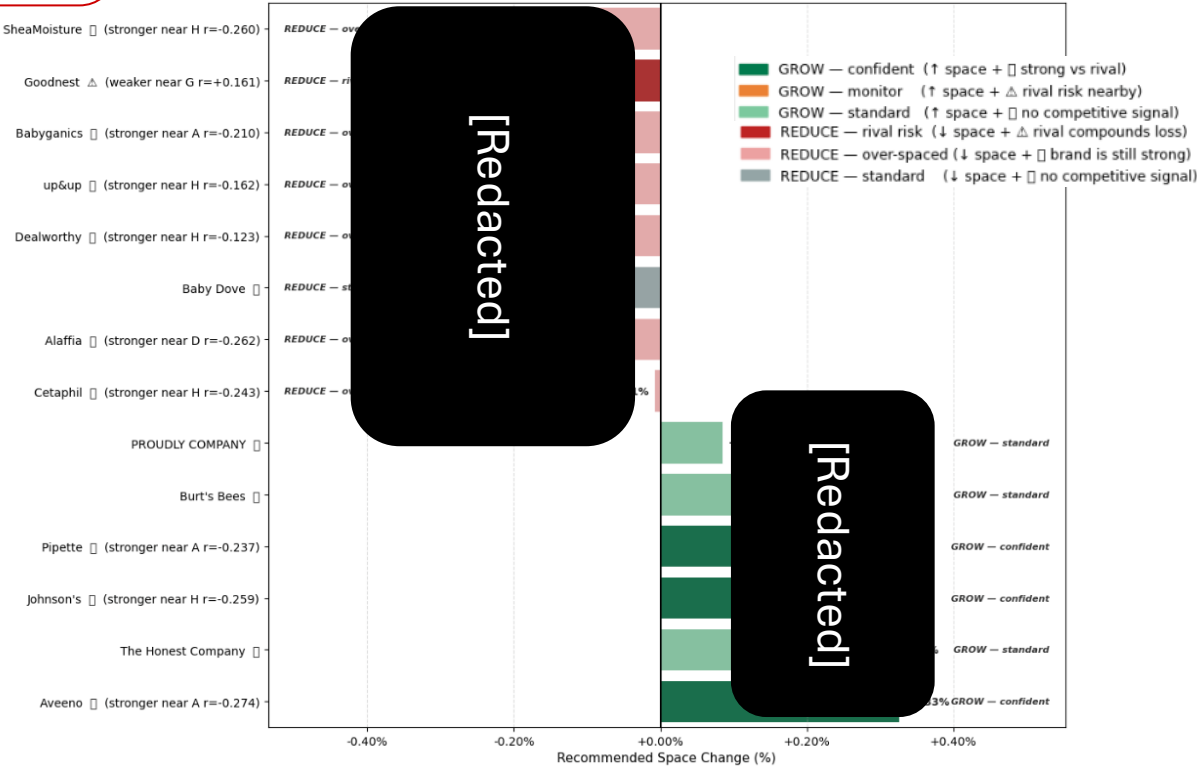
	RIVAL TAKES CUSTOMERS WHEN NEARBY			NEUTRAL		TARGET WINS — Stronger When Rival is Nearby		
	B	D	G	F	H	A	C	E
Assume	Online (e-comm) 800mi	Warehouse (Costco) 228mi	Specialty (Baby) 39mi	Mass Retail (Walmart) 1.6mi	General Retail 8.2mi	General Retail 9.2mi	Drug/Conv Retail 4.9mi	Grocery Chain 6.0mi
[Brand A] Gurobi: +[Redacted] GROW	 $r = +0.158$	 $r = -0.053$	 $r = +0.003$	 $r = -0.036$	 $r = -0.082$	 $r = -0.238$	 $r = -0.057$	 $r = -0.082$
[Brand B] Gurobi: +[Redacted] GROW	 $r = +0.101$	 $r = -0.050$	 $r = +0.004$	 $r = -0.037$	 $r = -0.053$	 $r = -0.203$	 $r = -0.028$	 $r = -0.053$
[Brand H] Gurobi: — MONITOR	 $r = +0.182$	 $r = +0.031$	 $r = -0.023$	 $r = +0.008$	 $r = -0.040$	 $r = -0.214$	 $r = -0.054$	 $r = -0.040$
[Brand I] Gurobi: +[Redacted] CAUTION	 $r = -0.011$	 $r = -0.037$	 $r = -0.052$	 $r = -0.018$	 $r = -0.137$	 $r = -0.249$	 $r = -0.139$	 $r = -0.137$
[Brand D] Gurobi: +[Redacted] GROW	 $r = +0.002$	 $r = +0.187$	 $r = +0.036$	 $r = -0.118$	 $r = -0.120$	 $r = -0.092$	 $r = -0.058$	 $r = -0.120$
[Brand E] Gurobi: — GROW	 $r = +0.071$	 $r = +0.266$	 $r = +0.180$	 $r = -0.079$	 $r = -0.042$	 $r = -0.143$	 $r = -0.056$	 $r = -0.042$
	3 brands weaker when B is nearby	2 brands weaker when D is nearby	1 [Brand E] weaker when G is nearby	1 [Brand D] otherwise neutral	2 brands stronger when H is nearby	5 brands stronger when A is nearby	1 [Brand I] stronger when C is nearby	2 brands stronger when E is nearby

Recommendations

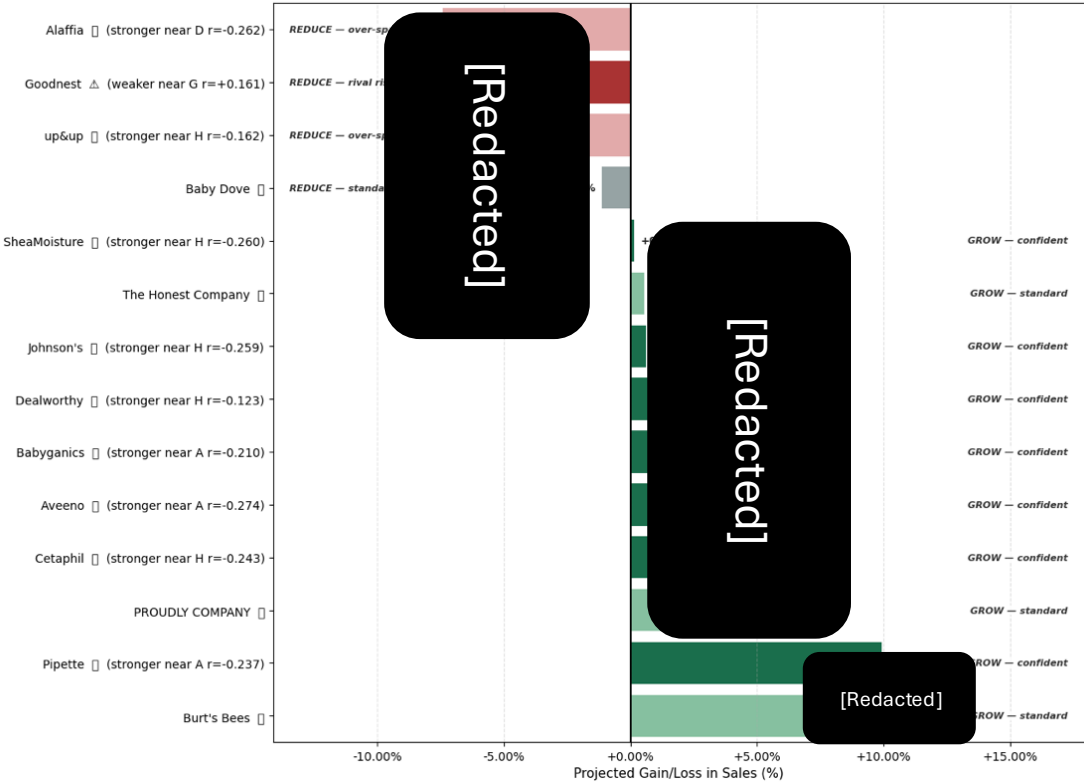


Example

2. Shelf Space Shift: Shampoos and Conditioners | Total: \$34.56M
Color = action zone | Italic label = combined recommendation



5. Net Revenue Impact: Shampoos and Conditioners | Total: \$34.56M
Color = action zone | Italic label = combined recommendation



Product	Category	Current Space (%)	New Space (%)	Recommended Shift (%)	Sales Growth (\$)	Sales Growth (%)	Corr (r)	Competition Signal	Final Action
[Brand F]+Shampoos_and_Conditioners	Shampoos and Conditioners	[Redacted]	[Redacted]	[Redacted]	[Redacted]	[Redacted]	-0.274	stronger near A (r=-0.274)	GROW — optimizer+ beats rival nearby
[Brand J]+Shampoos_and_Conditioners	Shampoos and Conditioners	[Redacted]	[Redacted]	[Redacted]	[Redacted]	[Redacted]	-0.210	stronger near A (r=-0.210)	CAUTIONS — optimizer says cut but brand still strong vs rival
[Brand K]+Shampoos_and_Conditioners	Shampoos and Conditioners	[Redacted]	[Redacted]	[Redacted]	[Redacted]	[Redacted]	+0.161	weaker near G (r=+0.161)	REDUCE — optimizer+ rival compounds the loss